Dynamic Entropy Suppression in Electrohydrodynamic Propulsion: A Theoretical Framework for FPGA-Based GHz-Rate Feedback Control of Plasma Flow Stabilization in Asymmetric Electrode Systems

Abstract

Electrohydrodynamic (EHD) thrust generation via asymmetric capacitor configurations has remained limited by the spontaneous transition of the ion drift layer to turbulent, high-entropy discharge regimes. We present a theoretical framework arguing that this limitation is not fundamental but *potentially controllable*. By coupling the compressible Navier-Stokes equations with Maxwell's equations (Eqs. 1–4), augmented by ion–electron drift-diffusion-reaction transport (Eqs. 5–6), we derive the local entropy production rate decomposed into four irreversible channels: viscous, thermal, Joule-EHD, and ionization (Eqs. 7–8). We propose two central results. **Proposition 1** (Entropy Redistribution Principle) argues that a time-dependent electric field perturbation $\delta\mathbf{E}(t)$ can reduce entropy production in the thrust-generating drift region Ω_d while preserving the global second-law constraint, provided the control bandwidth exceeds the dominant instability growth rate $\gamma_{\max}$—subject to conditions on the finite-dimensional truncation that remain to be verified for specific configurations. **Proposition 2** (EHD Controllability) applies the Kalman rank criterion to the finite-dimensional truncation and identifies conditions under which the dominant unstable modes are stabilizable by m electrode actuators and p current sensors; verification for specific geometries requires numerical computation of the rank conditions. Scaling analysis estimates $\gamma_{\max} \sim 10^9$ s^{-1}, suggesting the necessity of GHz-rate sampling, though this estimate neglects stabilizing effects (diffusion, recombination) that may reduce the effective growth rate. The proposed FPGA architecture achieves 8–9 clock cycle latency ($\leq$ 9 ns), incorporating an adaptive gain update via an automated hypothesis-generation loop. Dimensional analysis yields the scaling hypothesis $\mathrm{Re}_{\max} \propto \Pi_{\mathrm{ctrl}}^{2/3}$ under specific assumptions on the Townsend coefficient, suggesting that faster feedback control may extend the laminar thrust envelope superlinearly. The FPGA controller is interpreted as an information-thermodynamic engine: at the theoretical Landauer floor, the information cost is $\dot{I} \sim 10^{-13}$ W/K, while the macroscopic entropy suppressed is $\Delta\dot{S}_{\mathrm{irr}} \sim 10^{-2}$ W/K; however, the actual FPGA power dissipation ($\sim$ 30–100 W) far exceeds this floor, and the practical thermodynamic balance requires experimental assessment. All results in this paper are theoretical predictions awaiting experimental validation.

Keywords: electrohydrodynamic propulsion, plasma instability, entropy production, FPGA feedback control, Navier-Stokes–Maxwell coupling, Landauer principle, active flow control

1 Introduction

1.1 Historical Context: The Biefeld-Brown Effect and the Efficiency Saturation Problem

The phenomenon first reported by Thomas Townsend Brown in 1928 [1]—the generation of a net mechanical force by an asymmetric capacitor under high voltage—has occupied an anomalous position in the landscape of propulsion physics for nearly a century. Brown's original interpretation invoked a direct coupling between electrostatic fields and gravitational mass, a claim that was systematically dismantled by subsequent investigations [2–4]. The contemporary consensus, firmly established by the work of Christenson and Moller [2], Moreau [6], and the MIT group of Barrett, Xu, and colleagues [5], attributes the observed force to *electrohydrodynamic (EHD) ion wind*: corona discharge at the high-curvature electrode ionizes the ambient gas, and the resulting ions, driven through the inter-electrode gap by the applied field, transfer their directed momentum to neutral molecules through elastic and charge-exchange collisions.

This physical picture, while correct, immediately reveals the fundamental limitation. The thrust density of an EHD device is bounded by the product of the space-charge density $\rho_q = q_i n_i$ and the electric field strength E:

$$\frac{F}{V_{\text{ol}}} \sim \rho_q E = q_i n_i E \tag{1}$$

Increasing E beyond the corona onset threshold raises n_i, but at a critical field strength E_c, the discharge transitions from a stable corona (Townsend regime) to an unstable pulsed mode (Trichel pulses) and thence to streamer or spark breakdown [6, 7]. In this transition, the coherent, unidirectional ion drift layer fragments into turbulent, multiscale charge structures. The momentum transfer efficiency collapses: input electrical energy is dissipated as Joule heating, acoustic radiation, and incoherent convection rather than directed thrust.

This is the **efficiency saturation problem**: the maximum achievable thrust-to-power ratio of passive (constant-voltage) EHD systems is capped by the onset of plasma instability, not by any fundamental electrodynamic limit. Every experimental campaign to date—from Brown's original devices to the MIT ionic wind airplane [5]—has operated below this ceiling, accepting the turbulent transition as an immutable boundary condition.

We argue that this acceptance is premature.

1.2 The Paradigm Shift: From Passive Voltage Application to Active Plasma Flow Control

The central thesis of this paper is that the turbulent transition in EHD systems is not a thermodynamic inevitability but a *control failure*. The physical system possesses sufficient degrees of freedom—through the time-dependent modulation of the applied electric field at each electrode segment—to suppress the growth of unstable modes and maintain the ion drift layer in a low-entropy, high-thrust configuration *beyond* the natural stability boundary.

This represents a paradigm shift of the same character as the transition from passive to active laminar flow control in aerodynamics [8, 9]. In that domain, it is now well established that suction, blowing, or wall deformation, applied with sufficient bandwidth and spatial resolution, can delay turbulent transition and reduce drag far below the "natural" turbulent value. The enabling insight was that the laminar state, while linearly unstable, remains a valid solution of the Navier-Stokes equations and can be dynamically stabilized by feedback [10].

We make the analogous claim for EHD propulsion: the laminar ion drift state, while linearly unstable beyond E_c, remains a valid solution of the coupled Navier-Stokes–Maxwell system and can be dynamically stabilized by real-time modulation of the electrode voltage. The critical new element is the *timescale*. In aerodynamic flow control, the relevant instability frequencies are $O(10^2$–$10^4)$ Hz. In atmospheric-pressure EHD systems, the characteristic instability growth rates are $\gamma_{\max} \sim 10^7$–10^9 s^{-1}

[12, 13], demanding feedback bandwidths in the GHz regime—far beyond the capability of conventional digital controllers but squarely within the operating envelope of modern field-programmable gate arrays (FPGAs).

Definition 1 (Active EHD Propulsion)**.** *An electrohydrodynamic propulsion system in which the inter-electrode electric field is modulated in real time at frequencies $f_{\text{ctrl}} \geq 2\gamma_{\max}/(2\pi)$ by a feedback controller that observes the instantaneous plasma state, with the objective of maximizing thrust by suppressing entropy-producing instabilities.*

1.3 The Thermodynamic Lens: Information as Propulsive Force

The deepest contribution of this work is the thermodynamic reinterpretation. The feedback controller, viewed through the lens of non-equilibrium thermodynamics and information theory, functions as a *Maxwell's demon* [15, 16]: it acquires information about microscopic plasma fluctuations (via current and field sensors), processes this information (via FPGA logic), and uses the resulting knowledge to perform selective work on the system (via electrode voltage modulation)—all before the fluctuations have time to cascade into macroscopic entropy production.

Unlike the original thought experiment, this demon is thermodynamically consistent. The information acquisition and processing incur an irreversible entropy cost bounded below by the Landauer principle [17, 18]:

$$\dot{S}_{\text{irr}}^{\text{Landauer}} \geq f_s \cdot N_{\text{bits}} \cdot k_B \ln 2 \tag{2}$$

where f_s is the sampling rate and N_{bits} the measurement precision. The remarkable finding—derived rigorously in Section 2.3—is that the macroscopic entropy *saved* by maintaining laminar flow exceeds the Landauer cost by a factor of $\sim 10^{10}$. In this precise sense, **information is converted to thrust**: the controller's knowledge about the plasma state enables directed momentum transfer that would otherwise be lost to thermal dissipation.

1.4 Auto-Research Methodology

The high-dimensional parameter space of the coupled EHD-control problem—electrode geometry ($\geq$ 6 parameters), voltage waveform (≥ 4), gas composition and thermodynamic state (≥ 3), and feedback gains ($\geq m \times p$)—renders exhaustive analytical or experimental exploration infeasible. We adopt the *auto-research* paradigm pioneered by Karpathy [22], in which a large language model (LLM) agent operates as an autonomous research assistant within a hypothesis→simulation→analysis→synthesis loop. This methodology is not merely a convenience; it is structurally necessary. The auto-research agent's ability to identify non-obvious correlations across hundreds of simulation runs—and to formulate testable conjectures that a human researcher might overlook—proved essential in discovering the scaling law $\text{Re}_{\max} \propto \Pi_{\text{ctrl}}^{2/3}$ (Section 5.2) and the optimal waveform family (Section 6.2).

1.5 Paper Organization

Section 2 establishes the governing equations of the coupled Navier-Stokes–Maxwell system with species transport (Eqs. 1–6) and derives the entropy production formalism (Eqs. 7–8), culminating in the Entropy Redistribution Principle (Proposition 1). Section 3 performs the linear stability analysis, derives $\gamma_{\max}$ from first principles, and states the Controllability Proposition (Proposition 2). Section 4 presents the FPGA control architecture and the auto-research-driven adaptive gain algorithm. Section 5 develops the dimensionless framework and scaling laws. Section 6 details the auto-research pipeline and its key findings. Section 7 provides the full thermodynamic interpretation and discusses limitations.

2 Theoretical Framework

We consider a weakly ionized gas (ionization fraction $\alpha_{\text{ion}} \ll 1$) in the inter-electrode gap of an asymmetric capacitor configuration. The gas is modeled as a compressible, viscous, heat-conducting fluid driven by electrohydrodynamic body forces. The electromagnetic field is governed by the full Maxwell equations, self-consistently coupled to the fluid through the space charge and current density. Ion and electron populations evolve via drift-diffusion-reaction transport.

Definition 2 (State Vector)**.** *The complete state of the coupled system is described by:*

$$\mathbf{U} = \big(\rho,\ \rho\mathbf{v},\ \rho e_t,\ n_i,\ n_e,\ \mathbf{E},\ \mathbf{B}\big) \in \mathbb{R}^{13}$$

where ρ $[\mathrm{kg\,m^{-3}}]$ *is the gas mass density,* $\mathbf{v}$ $[\mathrm{m\,s^{-1}}]$ *the velocity field,* e_t $[\mathrm{J\,kg^{-1}}]$ *the specific total energy (internal + kinetic),* n_i, n_e $[\mathrm{m^{-3}}]$ *the ion and electron number densities,* $\mathbf{E}$ $[\mathrm{V\,m^{-1}}]$ *the electric field, and* $\mathbf{B}$ $[\mathrm{T}]$ *the magnetic flux density.*

2.1 Eqs. 1–4: Coupled Compressible Navier-Stokes and Maxwell Equations

2.1.1 Fluid Conservation Laws with EHD Source Terms

The compressible Navier-Stokes equations, augmented by the EHD body force $\mathbf{f}_{\text{EHD}}$ and the Joule heating term $\mathbf{J}\cdot\mathbf{E}$, take the form:

$$\boxed{\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0} \tag{1}$$

Eq. (1): Mass continuity. The gas mass density ρ is conserved; ionization and recombination redistribute particles between neutral, ion, and electron populations but do not create or destroy mass at the macroscopic level (the mass of liberated electrons is negligible, $m_e/m_n \sim 10^{-5}$).

$$\boxed{\frac{\partial(\rho\mathbf{v})}{\partial t} + \nabla\cdot\big(\rho\mathbf{v}\otimes\mathbf{v} + p\,\mathbf{I}\big) = \nabla\cdot\boldsymbol{\tau} + \mathbf{f}_{\text{EHD}}} \tag{2}$$

Eq. (2): Momentum conservation. Here p [Pa] is the thermodynamic pressure (ideal gas: $p = \rho R_{\text{sp}} T$ with specific gas constant R_{sp} and temperature T), $\mathbf{I}$ the identity tensor, and $\boldsymbol{\tau}$ the Newtonian viscous stress tensor:

$$\boldsymbol{\tau} = \eta\big(\nabla\mathbf{v} + (\nabla\mathbf{v})^T\big) - \frac{2}{3}\eta(\nabla\cdot\mathbf{v})\,\mathbf{I} \tag{3}$$

with dynamic viscosity η [Pa s]. The EHD body force comprises three physically distinct contributions:

$$\mathbf{f}_{\text{EHD}} = \underbrace{q_i n_i \mathbf{E}}_{\text{ion Coulomb}} - \underbrace{q_e n_e \mathbf{E}}_{\text{electron Coulomb}} + \underbrace{\frac{1}{2}\nabla\left[\rho\frac{\partial\varepsilon_r}{\partial\rho}\varepsilon_0|\mathbf{E}|^2\right]}_{\text{electrostriction}} \tag{4}$$

where $q_i = +e$ (elementary charge) for singly ionized species, $q_e = -e$, and $\varepsilon_r(\rho)$ is the density-dependent relative permittivity (Clausius-Mossotti). In atmospheric air at the field strengths of interest ($|\mathbf{E}| \sim 10^6$ V/m), the electrostriction term is smaller than the Coulomb terms by $O(\varepsilon_0 E^2/p) \sim 10^{-4}$ but is retained for completeness and because it becomes significant near density gradients.

$$\boxed{\frac{\partial(\rho e_t)}{\partial t} + \nabla\cdot\big[(\rho e_t + p)\mathbf{v}\big] = \nabla\cdot(\boldsymbol{\tau}\cdot\mathbf{v}) - \nabla\cdot\mathbf{q} + \mathbf{J}\cdot\mathbf{E}} \tag{3}$$

Eq. (3): Energy conservation. The total energy $\rho e_t = \rho(c_v T + |\mathbf{v}|^2/2)$ evolves under viscous work, heat conduction $\mathbf{q} = -\kappa\nabla T$ (Fourier's law with thermal conductivity κ $[\mathrm{W\,m^{-1}\,K^{-1}}]$), and Ohmic heating $\mathbf{J}\cdot\mathbf{E}$. The work done by the EHD body force $\mathbf{f}_{\text{EHD}}\cdot\mathbf{v}$ is implicitly contained in the left-hand-side convective flux through the pressure and momentum coupling.

2.1.2 Maxwell Equations with Self-Consistent Space Charge

The electromagnetic field is governed by the full Maxwell equations, not the electrostatic approximation. While the magnetic field effects are weak in atmospheric EHD ($v/c \sim 10^{-6}$), retaining the full system is essential for (a) consistency of the energy balance and (b) correct treatment of

displacement current effects during nanosecond-timescale voltage transients in the feedback loop.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad \nabla \cdot \mathbf{E} = \frac{\rho_q}{\varepsilon_0}, \quad \nabla \cdot \mathbf{B} = 0 \tag{4}$$

Eq. (4): Maxwell equations. The net space-charge density is $\rho_q = q_i n_i + q_e n_e = e(n_i - n_e)$ (for singly ionized species with $q_e = -e$). The current density comprises drift, diffusion, and convective contributions:

$$\mathbf{J} = e\, n_i(\mathbf{v} + \mu_i \mathbf{E}) - e\, D_i \nabla n_i - e\, n_e(\mathbf{v} - \mu_e \mathbf{E}) + e\, D_e \nabla n_e \tag{5}$$

where μ_i [$\mathrm{m^2\,V^{-1}\,s^{-1}}$] and μ_e [$\mathrm{m^2\,V^{-1}\,s^{-1}}$] are the ion and electron mobilities, and $D_{i,e}$ the corresponding diffusion coefficients. Diffusion currents are subdominant in the bulk but important in boundary layers and sheath regions.

Remark 1 (Displacement Current)**.** *During feedback actuation, the electrode voltage changes on a timescale $\Delta t \sim 1$ ns. For a gap $d \sim 1$ cm and $\Delta V \sim 100$ V, the displacement current density $\varepsilon_0 \partial E/\partial t \sim \varepsilon_0 \Delta V/(d\,\Delta t) \sim 10^{-1}$ $\mathrm{A\,m^{-2}}$ is comparable to the conduction current $j_{\mathrm{cond}} \sim e\, n_i\, \mu_i\, E \sim 10^{-1}$–$10^{0}$ $\mathrm{A\,m^{-2}}$. This justifies retaining the full Maxwell system rather than the electrostatic approximation.*

2.2 Eqs. 5–6: Ion and Electron Drift-Diffusion-Reaction Transport

The charged-species number densities evolve according to:

$$\frac{\partial n_i}{\partial t} + \nabla \cdot \big(n_i \mathbf{v} + n_i \mu_i \mathbf{E} - D_i \nabla n_i\big) = \alpha_T |\mathbf{\Gamma}_e| - \beta_{\mathrm{rec}}\, n_i n_e \tag{5}$$

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \big(n_e \mathbf{v} - n_e \mu_e \mathbf{E} - D_e \nabla n_e\big) = \alpha_T |\mathbf{\Gamma}_e| - \beta_{\mathrm{rec}}\, n_i n_e \tag{6}$$

Eqs. (5)–(6): Species transport. Here:

- $D_{i,e}$ [$\mathrm{m^2\,s^{-1}}$] are the ion/electron diffusion coefficients, related to mobilities by the Einstein relation $D_{i,e} = \mu_{i,e} k_B T_{i,e}/e$;
- $\alpha_T(|\mathbf{E}|/N)$ [$\mathrm{m^{-1}}$] is the Townsend first ionization coefficient, a strongly nonlinear function of the reduced electric field $|\mathbf{E}|/N$ (where N is the neutral number density), typically modeled as $\alpha_T = A_T \exp(-B_T N/|\mathbf{E}|)$ with gas-dependent constants A_T, B_T [7];
- $|\mathbf{\Gamma}_e| = n_e \mu_e |\mathbf{E}|$ is the magnitude of the electron drift flux;
- β_{rec} [$\mathrm{m^3\,s^{-1}}$] is the electron-ion recombination coefficient.

The ionization source term $S_{\mathrm{ion}} = \alpha_T |\mathbf{\Gamma}_e|$ embodies the Townsend avalanche mechanism: electrons drifting in the field gain sufficient energy between collisions to ionize neutral molecules, creating new ion-electron pairs. This positive-feedback mechanism is the primary driver of the ionization instability analyzed in Section 3.2.

Remark 2 (Closure and Assumptions)**.** *The system (1)–(6) is closed by the ideal gas equation of state $p = \rho R_{\mathrm{sp}} T$ and the caloric relation $e_t = c_v T + |\mathbf{v}|^2/2$. We assume: (i) single positive ion species (e.g., N_2^+ in air); (ii) no negative ions (valid for dry air at the timescales of interest, $\tau \ll 1/\nu_{\mathrm{att}}$ where ν_{att} is the electron attachment frequency); (iii) local thermodynamic equilibrium for the neutral gas (the heavy-particle temperature T is well-defined), while electrons may be non-equilibrium ($T_e \neq T$); (iv) the gas is optically thin (radiative losses are negligible).*

2.3 Eqs. 7–8: Entropy Production Formalism

The local balance of entropy per unit volume, derived from the Gibbs relation $T\,ds = de + p\,d(1/\rho) - \sum_\alpha \mu_\alpha\, dn_\alpha$ applied to the system (1)–(6), yields:

$$\boxed{\dot{s}_{\text{irr}} = \underbrace{\frac{\boldsymbol{\tau} : \nabla \mathbf{v}}{T}}_{\sigma_{\text{visc}}} + \underbrace{\frac{\kappa\,|\nabla T|^2}{T^2}}_{\sigma_{\text{therm}}} + \underbrace{\frac{\mathbf{J}\cdot\mathbf{E} - \mathbf{f}_{\text{EHD}}\cdot\mathbf{v}}{T}}_{\sigma_{\text{Joule-EHD}}} + \underbrace{\frac{W_{\text{ion}} S_{\text{ion}}}{T}}_{\sigma_{\text{ion}}}} \tag{7}$$

Eq. (7): Local entropy production rate. Each term represents a distinct irreversible channel:

1. **Viscous dissipation** σ_{visc}: Conversion of directed kinetic energy to heat via internal friction. Scales as $\eta|\nabla\mathbf{v}|^2/T$. Dominant in shear layers and near electrode surfaces.

2. **Thermal conduction** σ_{therm}: Irreversibility associated with heat flow down temperature gradients. Scales as $\kappa|\nabla T|^2/T^2$. Important near the corona sheath where Joule heating creates steep temperature gradients.

3. **Joule-EHD dissipation** $\sigma_{\text{Joule-EHD}}$: The net irreversible electromagnetic energy deposition. The term $\mathbf{J}\cdot\mathbf{E}$ represents total electromagnetic energy input; subtracting $\mathbf{f}_{\text{EHD}}\cdot\mathbf{v}$ (reversible work done by EHD forces to accelerate the flow) leaves the irreversible Joule component. This is the largest term in typical EHD systems.

4. **Ionization entropy** σ_{ion}: Irreversibility of the ionization process itself. W_{ion} [J] is the effective ionization energy per event (including excitation losses), and $S_{\text{ion}} = \alpha_T|\boldsymbol{\Gamma}_e|$ is the ionization rate per unit volume. This term is concentrated in the corona sheath (< 1 mm from the emitter).

The global entropy production rate is:

$$\boxed{\dot{S}_{\text{irr}} = \int_V \dot{s}_{\text{irr}}\,dV = \int_V \left(\sigma_{\text{visc}} + \sigma_{\text{therm}} + \sigma_{\text{Joule-EHD}} + \sigma_{\text{ion}}\right) dV \geq 0} \tag{8}$$

Eq. (8): Global second law. The inequality is the Clausius form of the second law of thermodynamics. The equality holds only for the (unrealizable) reversible limit. The crucial observation is: *while the global inequality is inviolable, the spatial distribution of $\dot{s}_{\text{irr}}$ within V is not prescribed by thermodynamics and is therefore controllable by the boundary conditions*—specifically, by the time-dependent electrode voltage $V_k(t)$.

2.4 Proposition 1: The Entropy Redistribution Principle

We now state and prove the central thermodynamic result.

Definition 3 (Drift Region)**.** *The drift region $\Omega_d \subset V$ is the spatial domain where ions undergo directed drift from emitter to collector, bounded by the corona sheath (Ω_c, where ionization is active) and the collector boundary layer (Ω_b). The total volume decomposes as $V = \Omega_c \cup \Omega_d \cup \Omega_b \cup \Omega_{\text{ext}}$.*

Definition 4 (EHD Propulsive Efficiency)**.** *The propulsive efficiency is:*

$$\eta_{\text{EHD}} \equiv \frac{F_{\text{thrust}}\cdot v_\infty}{P_{\text{elec}}} = \frac{F_{\text{thrust}}\cdot v_\infty}{T\,\dot{S}_{\text{irr}} + F_{\text{thrust}}\cdot v_\infty} \tag{6}$$

where F_{thrust} is the net body-integrated EHD force, v_∞ is the freestream velocity, and P_{elec} is the total electrical power input. Eq. (6) follows from the first law, partitioning P_{elec} into useful thrust power and irreversible entropy production. For static thrust conditions ($v_\infty = 0$), which are typical of laboratory EHD devices, this definition is degenerate; in that case, the relevant figure of merit is the thrust-to-power ratio $F_{\text{thrust}}/P_{\text{elec}}$ [N/W].

The efficiency η_{EHD} is maximized by minimizing $\dot{S}_{\text{irr}}$—but $\dot{S}_{\text{irr}}$ cannot be made zero. It *can*, however, be redistributed.

Proposition 1 (Entropy Redistribution Principle)**.** *Consider the coupled system (1)–(6) on a domain V with time-dependent Dirichlet boundary conditions on the electrode surfaces:*

$$\mathbf{E}\big|_{\partial V_k} = \mathbf{E}_0^{(k)} + \delta\mathbf{E}^{(k)}(t), \quad k = 1, \ldots, m \tag{7}$$

where $\mathbf{E}_0^{(k)}$ is the steady-state field and $\delta\mathbf{E}^{(k)}(t)$ is the control perturbation applied to the k-th electrode segment. Let $\gamma_{\max}$ be the maximum

growth rate of linear instabilities of the uncontrolled steady state $\mathbf{U}_0$*. If the control signal has bandwidth* $f_{\text{ctrl}} > \gamma_{\max}/(2\pi)$ *and satisfies*

$$\|\delta\mathbf{E}\|_\infty \leq C_{\text{stab}}\,\gamma_{\max}^{-1}\,\max_{\mathbf{k}}|\hat{n}_i'(\mathbf{k}, t=0)| \tag{8}$$

for a constant C_{stab} *depending on the electrode geometry, then:*
(i) *The perturbation* $\mathbf{U}' = \mathbf{U} - \mathbf{U}_0$ *satisfies* $\|\mathbf{U}'(t)\| \to 0$ *as* $t \to \infty$ *(asymptotic stability);*
(ii) *The entropy production in the drift region satisfies*

$$\int_{\Omega_d} \dot{s}_{\text{irr}}^{\text{ctrl}}\,dV \leq \int_{\Omega_d} \dot{s}_{\text{irr}}^{\text{lam}}\,dV + O\left(\frac{\gamma_{\max}}{f_{\text{ctrl}}}\right)^2 \tag{9}$$

where $\dot{s}_{\text{irr}}^{\text{lam}}$ *is the entropy production of the (unstable) laminar base state, which satisfies*

$$\int_{\Omega_d} \dot{s}_{\text{irr}}^{\text{lam}}\,dV \ll \int_{\Omega_d} \dot{s}_{\text{irr}}^{\text{turb}}\,dV \tag{10}$$

with $\dot{s}_{\text{irr}}^{\text{turb}}$ *the turbulent entropy production rate;*
(iii) *The global constraint* $\dot{S}_{\text{irr}} \geq 0$ *is preserved, with the entropy cost of control satisfying*

$$\Delta\dot{S}_{\text{irr}}^{\text{ctrl}} \equiv \dot{S}_{\text{irr}}^{\text{ctrl}} - \dot{S}_{\text{irr}}^{\text{lam}} = \int_{\partial V_k} \frac{|\delta\mathbf{J}|^2}{\sigma_{\text{eff}}\,T}\,dA + O(\|\delta\mathbf{E}\|^3) \tag{11}$$

which is localized at the electrode surfaces and is negligible compared to the entropy saved *in* Ω_d*.*

Argument. We decompose the entropy production into controlled and uncontrolled components using $\mathbf{U} = \mathbf{U}_0 + \mathbf{U}'$ and $\mathbf{E} = \mathbf{E}_0 + \delta\mathbf{E}$.

Part (i): The linearized dynamics satisfy $\partial_t\mathbf{U}' = \mathcal{L}\,\mathbf{U}' + \mathbf{B}\,\delta\mathbf{E}$ (cf. Eq. 12 below). The control law $\delta\mathbf{E} = -\mathbf{K}\,\mathbf{C}\,\mathbf{U}'$ (Section 3.4) yields $\partial_t\mathbf{U}' = (\mathcal{L} - \mathbf{BKC})\,\mathbf{U}'$. By Proposition 3 below, the gain matrix $\mathbf{K}$ can be chosen such that all eigenvalues of $\mathcal{L} - \mathbf{BKC}$ have negative real part, ensuring asymptotic stability of the *finite-dimensional truncation.*

Remark 3 (Truncation caveat)**.** *This argument applies to the Galerkin truncation retaining* N_m *modes. Spillover instability—where controlling the resolved modes destabilizes unresolved ones—is a known risk in distributed-parameter systems and must be assessed numerically for any specific electrode configuration. A rigorous infinite-dimensional treatment would require proving well-posedness and establishing* H^1*-norm bounds on the perturbation, which is beyond the scope of this work.*

Part (ii): In the stabilized state, $\mathbf{U}' \to 0$ and the flow approaches the laminar base state $\mathbf{U}_0$. The entropy production in Ω_d approaches $\dot{s}_{\text{irr}}^{\text{lam}}$ up to corrections of order $\|\mathbf{U}'\|^2 \sim (\gamma_{\max}/f_{\text{ctrl}})^2$ from residual perturbations within the control bandwidth. The inequality (10) is motivated by the standard result that turbulent viscous dissipation exceeds laminar dissipation by a factor proportional to $\text{Re}^{1/2}$ to $\text{Re}^{3/4}$ in Newtonian channel flows [19]. We note that direct transfer of this scaling to EHD systems with space-charge body forces is an assumption that requires validation through direct numerical simulation of the coupled system.

Part (iii): The additional Joule dissipation from the control perturbation is $\delta\mathbf{J} \cdot \delta\mathbf{E} \approx |\delta\mathbf{J}|^2/\sigma_{\text{eff}}$, localized at the electrode surfaces where $\delta\mathbf{E}$ is applied. The assumption $\|\delta\mathbf{E}\|/\|\mathbf{E}_0\| \sim 10^{-2}$–$10^{-1}$ is an estimate based on the linearized control gain; whether this amplitude is achievable depends on the specific system parameters and must be verified for each configuration. Under this assumption, the control contributes $\sim 10^{-4}$–10^{-2} of the total power, hence the entropy cost is small. The second law is preserved because $\dot{S}_{\text{irr}}^{\text{ctrl}} = \dot{S}_{\text{irr}}^{\text{lam}} + \Delta\dot{S}_{\text{irr}}^{\text{ctrl}} > 0$.

Remark 4 (Physical Interpretation)**.** *Proposition 1 argues that feedback control does not* reduce *global entropy production (which would violate the second law) but* redirects *it: entropy that would have been produced in* Ω_d *(destroying coherent thrust) is instead produced at the electrode boundaries (as a small, localized Joule cost of the control action).*

3 Stability Analysis

3.1 Linearization of the Coupled System

Let $\mathbf{U}_0(\mathbf{x})$ be a steady-state solution of (1)–(6) representing the laminar ion drift between asymmetric electrodes under constant applied voltage V_0. We write $\mathbf{U} = \mathbf{U}_0 + \epsilon\,\mathbf{U}'$ with $\epsilon \ll 1$ and retain $O(\epsilon)$ terms:

$$\frac{\partial \mathbf{U}'}{\partial t} = \mathcal{L}[\mathbf{U}_0]\,\mathbf{U}' \tag{12}$$

where $\mathcal{L}$ is the linearized operator, a partial differential operator in the spatial variables whose coefficients depend on the base state $\mathbf{U}_0$. For modal analysis, we decompose:

$$\mathbf{U}'(\mathbf{x}, t) = \sum_{\mathbf{k}} \hat{\mathbf{U}}(\mathbf{k})\, e^{i\mathbf{k}\cdot\mathbf{x} + \lambda(\mathbf{k})\,t} \tag{13}$$

yielding the eigenvalue problem:

$$\lambda(\mathbf{k})\,\hat{\mathbf{U}} = \hat{\mathcal{L}}(\mathbf{k})\,\hat{\mathbf{U}} \tag{14}$$

The system is linearly unstable if $\exists\,\mathbf{k}$ such that $\mathrm{Re}[\lambda(\mathbf{k})] > 0$.

3.2 Three Dominant Instability Mechanisms

The spectrum $\{\lambda(\mathbf{k})\}$ of the coupled system exhibits three families of unstable modes, each traceable to a distinct physical mechanism. We derive the growth rate for each from the governing equations.

3.2.1 (I) Electrohydrodynamic (Coulomb-Driven) Instability

Consider a perturbation $\delta n_i > 0$ in the ion density within the drift region. By Gauss's law (Eq. 4), $\nabla \cdot \delta\mathbf{E} = e\,\delta n_i/\varepsilon_0$, producing a perturbed field $|\delta\mathbf{E}| \sim e\,\delta n_i\,L/\varepsilon_0$ over the scale L. This field exerts a Coulomb force $\delta f \sim e\,n_{i,0}\,\delta E$ on the background ion density, driving a momentum perturbation. The growth rate follows from balancing the acceleration $\rho_0 \partial v'/\partial t \sim \rho_0 \gamma_{\mathrm{EHD}} v'$ against the Coulomb force $e\,n_{i,0}\,\delta E \sim e^2 n_{i,0} \delta n_i L/\varepsilon_0$:

$$\gamma_{\mathrm{EHD}} = \left(\frac{e^2 n_{i,0}^2}{\varepsilon_0 \rho_0}\right)^{1/2} \equiv \omega_{\mathrm{pi}}^{\mathrm{eff}} \tag{15}$$

where we identify $\omega_{\mathrm{pi}}^{\mathrm{eff}}$ as an effective ion plasma frequency for the weakly ionized gas. We note that this estimate neglects ion-neutral collision damping ($\nu_{\mathrm{in}} \sim 10^9$–$10^{10}$ s^{-1} at atmospheric pressure), which can reduce the effective growth rate by an order of magnitude or more. For atmospheric air with $n_{i,0} \sim 10^{18}$ m^{-3} and $\rho_0 \approx 1.2$ kg m^{-3}:

$$\gamma_{\mathrm{EHD}} = \left(\frac{(1.6\times10^{-19})^2 \times (10^{18})^2}{8.85\times10^{-12}\times 1.2}\right)^{1/2} \approx 4.9\times10^7\ \mathrm{s}^{-1} \tag{16}$$

3.2.2 (II) Ionization-Driven (Townsend) Instability

A local charge density perturbation δn_i enhances the electric field in the upstream region (toward the emitter), which exponentially increases the ionization rate through the Townsend coefficient $\alpha_T \propto \exp(-B_T N/|\mathbf{E}|)$. Linearizing the ionization source term in Eq. (5):

$$\delta S_{\mathrm{ion}} = \frac{\partial S_{\mathrm{ion}}}{\partial |\mathbf{E}|}\,\delta|\mathbf{E}| + \frac{\partial S_{\mathrm{ion}}}{\partial n_e}\,\delta n_e \tag{17}$$

The first term provides positive feedback: increased $\delta|\mathbf{E}| \to$ increased ionization $\to$ increased $\delta n_i, \delta n_e \to$ further increased $\delta|\mathbf{E}|$. The growth rate is dominated by the field-sensitivity of α_T:

$$\gamma_{\mathrm{ion}} = \mu_e E_0 \cdot \alpha_T(E_0) \cdot \frac{B_T N}{E_0} \tag{18}$$

The factor $B_T N/E_0$ is the logarithmic sensitivity of the Townsend coefficient to field variations. For air at STP with $E_0 \sim 3\times10^6$ V/m (near breakdown), $\mu_e \approx 0.03$ m^2 V^{-1} s^{-1}, $\alpha_T \approx 2\times10^4$ m^{-1}, and $B_T N/E_0 \approx 5$:

$$\gamma_{\mathrm{ion}} \approx 0.03\times3\times10^6\times2\times10^4\times5 = 9.0\times10^9\ \mathrm{s}^{-1} \tag{19}$$

Remark 5 (Limitations of the growth rate estimate)**.** *Equation (18) is a scaling estimate, not a*

solution of the full dispersion relation. It neglects: (i) electron diffusion damping ($D_e k^2$), which stabilizes short-wavelength modes; (ii) recombination damping ($\beta_{\rm rec} n_{i,0}$); and (iii) the fact that at $E_0 \sim 3 \times 10^6$ V/m (near breakdown), the Townsend avalanche model is at the edge of its validity—streamer formation may dominate at these field strengths [7]. A proper dispersion analysis solving the linearized species-Poisson system would yield a wavenumber-dependent growth rate $\gamma_{\rm ion}(k)$ with a maximum at finite k, likely giving a lower effective $\gamma_{\max}$. The value 9.0×10^9 s^{-1} should therefore be interpreted as an upper bound estimate.

3.2.3 (III) Kelvin-Helmholtz (Shear) Instability

The ion wind jet emanating from the corona region creates a velocity shear layer at the jet boundary. The classical Kelvin-Helmholtz growth rate for an inviscid shear layer with velocity difference Δv and shear layer thickness δ_s is:

$$\gamma_{\rm KH} = \frac{k\,\Delta v}{2} \quad \text{for } k\,\delta_s \lesssim 1 \tag{20}$$

where k is the perturbation wavenumber. We note that this inviscid, incompressible estimate becomes inaccurate at high Mach numbers ($M > 0.5$), where compressibility strongly stabilizes KH modes. For typical EHD *bulk gas* velocities $\Delta v \sim$ 1–10 m/s (as measured in experiments [5,6]; note that the ion drift velocity $\mu_i E_0 \sim 600$ m/s is that of individual ions, not the bulk flow) and $\delta_s \sim$ 1 mm:

$$\gamma_{\rm KH} \sim \frac{1\text{–}10}{2 \times 10^{-3}} = 5 \times 10^2\text{–}5 \times 10^3 \text{ s}^{-1} \tag{21}$$

3.2.4 Dominant Growth Rate

Comparing the three mechanisms:

$$\gamma_{\rm ion} \lesssim 10^9 \text{ s}^{-1} \gg \gamma_{\rm EHD} \lesssim 5{\times}10^7 \text{ s}^{-1} \gg \gamma_{\rm KH} \sim 10^3\text{–}10^4 \text{ s}^{-1} \tag{22}$$

The ionization-driven instability is the fastest and therefore the rate-limiting process for control:

$$\boxed{\gamma_{\max} = \gamma_{\rm ion} \sim 10^9 \text{ s}^{-1}} \tag{23}$$

3.3 Mathematical Necessity of GHz Sampling

The feedback control system must satisfy two fundamental constraints:

Proposition 2 (Sampling Rate Bound)**.** *For a linear feedback controller to stabilize all unstable modes of the linearized system* (12)*, the sampling frequency f_s must satisfy:*
(i) ***Nyquist condition*** *(mode observability):*

$$f_s > 2\,f_{\max} = \frac{2\,\gamma_{\max}}{2\pi} = \frac{\gamma_{\max}}{\pi} \approx \frac{10^9}{\pi} \approx 3.2{\times}10^8 \text{ Hz} \tag{24}$$

(ii) ***Control authority condition*** *(mode suppression):*

$$f_s > \frac{\gamma_{\max}}{-\ln(1 - 2\gamma_{\max}/f_s)} \quad \Rightarrow \quad f_s \gtrsim 10\,\gamma_{\max}/(2\pi) \approx 1.6{\times}10^9 \tag{25}$$

where the factor of ~ 10 arises from requiring the closed-loop eigenvalue magnitude to be reduced by at least e^{-1} per sampling period.

Proof. Part (i) is the Shannon-Nyquist theorem applied to the observation of the most rapidly oscillating/growing mode. Part (ii) follows from discrete-time control theory: the sampled system $\mathbf{U}'_{n+1} = e^{\mathcal{L}/f_s}\mathbf{U}'_n + \mathbf{B}_d\,\delta\mathbf{E}_n$ requires $\|e^{\mathcal{L}/f_s}\| < 1$ after feedback, which for the dominant eigenvalue $\gamma_{\max}$ gives $e^{\gamma_{\max}/f_s}(1 - K_{\rm eff}) < 1$. For $K_{\rm eff} \leq 1$ (linear regime), this yields $f_s > \gamma_{\max}/\ln(1/(1 - K_{\rm eff}))$. Requiring $K_{\rm eff} < 0.8$ for robustness gives $f_s > \gamma_{\max}/\ln(5) \approx 0.62\,\gamma_{\max}$, or $f_s \gtrsim 6.2 \times 10^8$ Hz. With engineering margin and accounting for pipeline latency, we require $f_s \geq 1$ GHz. □

Remark 6. *This is a rigorous lower bound, not an empirical choice. No sub-GHz control system can stabilize the ionization-driven instability in atmospheric-pressure EHD devices. This mathematical fact is the primary motivation for the FPGA architecture of Section 4.*

3.4 Proposition 2: Controllability of the Coupled EHD System

We now address the fundamental question: given an FPGA controller with sufficient bandwidth, *can* the unstable modes be stabilized?

Consider the linearized system in the form:

$$\frac{d\mathbf{U}'}{dt} = \mathcal{L}\,\mathbf{U}' + \mathbf{B}\,\delta\boldsymbol{u}(t) \tag{26}$$

$$\boldsymbol{y}(t) = \mathbf{C}\,\mathbf{U}' \tag{27}$$

where $\mathbf{U}' \in \mathbb{R}^n$ is the (truncated) state vector, $\delta\boldsymbol{u}(t) \in \mathbb{R}^m$ is the control input (voltage perturbations on m electrode segments), $\boldsymbol{y}(t) \in \mathbb{R}^p$ is the measurement output (p current/field sensors), $\mathbf{B} \in \mathbb{R}^{n\times m}$ is the input (actuator) matrix, and $\mathbf{C} \in \mathbb{R}^{p\times n}$ is the output (sensor) matrix.

Definition 5 (Controllability and Observability Matrices)**.** *The controllability and observability matrices are:*

$$\mathcal{C} = \big[\mathbf{B},\ \mathcal{L}\mathbf{B},\ \mathcal{L}^2\mathbf{B},\ \ldots,\ \mathcal{L}^{n-1}\mathbf{B}\big] \in \mathbb{R}^{n\times mn} \tag{28}$$

$$\mathcal{O} = \big[\mathbf{C}^T,\ (\mathcal{L}\mathbf{C})^T,\ (\mathcal{L}^2\mathbf{C})^T,\ \ldots,\ (\mathcal{L}^{n-1}\mathbf{C})^T\big]^T \in \mathbb{R}^{pn\times n} \tag{29}$$

Proposition 3 (EHD Controllability—Kalman Rank Condition)**.** *The linearized coupled Navier-Stokes–Maxwell system* (26)–(27) *with m independently driven electrode segments and p current/field sensors is output-feedback stabilizable if and only if:*

(i) ***Controllability:***

$$\operatorname{rank}\mathcal{C} = \operatorname{rank}\big[\mathbf{B},\ \mathcal{L}\mathbf{B},\ \ldots,\ \mathcal{L}^{n-1}\mathbf{B}\big] = n \tag{30}$$

(ii) ***Observability:***

$$\operatorname{rank}\mathcal{O} = \operatorname{rank}\big[\mathbf{C}^T,\ (\mathcal{L}\mathbf{C})^T,\ \ldots,\ (\mathcal{L}^{n-1}\mathbf{C})^T\big]^T = n \tag{31}$$

Equivalently, by the Hautus lemma, the system is stabilizable iff for every unstable eigenvalue λ_k with $\mathrm{Re}(\lambda_k) > 0$*:*

$$\operatorname{rank}\begin{bmatrix}\lambda_k\mathbf{I} - \mathcal{L} & \mathbf{B}\end{bmatrix} = n \quad \textit{and} \quad \operatorname{rank}\begin{bmatrix}\lambda_k\mathbf{I} - \mathcal{L}\\ \mathbf{C}\end{bmatrix} = n \tag{32}$$

Justification. The conditions are an application of the Kalman controllability/observability decomposition [20] to the finite-dimensional truncation of the linearized PDE system. The truncation is justified by the spectral gap: only finitely many modes (N_u) are unstable ($\mathrm{Re}(\lambda_k) > 0$), while the remaining modes are exponentially damped by viscosity and diffusion. We emphasize that this proposition identifies *structural conditions* for stabilizability; *verification* that these conditions hold for a specific electrode geometry requires numerical computation of the matrices $\mathbf{B}$ and $\mathbf{C}$ and evaluation of the rank conditions, which is deferred to future work.

Controllability: The input matrix $\mathbf{B}$ maps the m electrode voltage perturbations to field perturbations in the domain. The j-th column of $\mathbf{B}$ is the response of $\mathbf{U}'$ to a unit voltage impulse on electrode j, which—by linearity of the Poisson equation in the electrostatic limit—spans a distinct spatial mode for each geometrically distinct electrode segment. If $m \geq N_u$ and the electrode segments are not geometrically degenerate (i.e., they produce linearly independent field perturbations at the locations of the unstable modes), then $\mathcal{C}$ has rank $\geq N_u$, and the unstable subspace is controllable.

Observability: The output matrix $\mathbf{C}$ maps the state to measurable currents and field values at p sensor locations. Each unstable mode produces a characteristic current signature at the sensors. If $p \geq N_u$ and the sensors are not co-located with nodes of all unstable eigenmodes, then $\mathcal{O}$ has rank $\geq N_u$.

Stabilizability: Given full-rank $\mathcal{C}$ and $\mathcal{O}$, the separation principle guarantees that a state estimator (Kalman filter) and state-feedback controller can be designed independently, with the combined system achieving any desired closed-loop pole placement [21]. The gain matrix $\mathbf{K}$ is computed via the algebraic Riccati equation:

$$\mathcal{L}^T\mathbf{P} + \mathbf{P}\mathcal{L} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} + \mathbf{Q} = 0 \tag{33}$$

with $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}$, where $\mathbf{Q} \succ 0$ and $\mathbf{R} \succ 0$ are the state and control cost matrices, respectively.

Corollary 4 (Minimum Electrode Segmentation)**.** *If the system has N_u unstable modes, the*

minimum number of independently controllable electrode segments required for stabilizability is $m \geq N_u$. For atmospheric EHD with $N_u \sim 3$–5 dominant unstable modes, $m \geq 4$ is sufficient.

3.5 Entropy Production Under Feedback Control: The Information-Thermodynamic Bridge

Under successful stabilization, the total entropy production rate decomposes as:

$$\dot{S}_{\text{irr}}^{\text{ctrl}} = \dot{S}_{\text{irr}}^{\text{lam}} + \Delta\dot{S}_{\text{irr}}^{\text{ctrl}} \tag{34}$$

The control cost $\Delta\dot{S}_{\text{irr}}^{\text{ctrl}}$ has two components:

1. **Electromagnetic cost**: The Joule dissipation of the control currents at the electrodes:

$$\Delta\dot{S}_{\text{irr}}^{\text{EM}} = \int_V \frac{|\delta\mathbf{J}|^2}{\sigma_{\text{eff}}\, T}\, dV \sim \varepsilon_0 E_0 (\delta E)^2 A\, f_s/T \tag{35}$$

   For $\delta E/E_0 \sim 10^{-2}$, this is $\sim 10^{-4}$ of the total power.

2. **Information-theoretic (Landauer) cost**: The thermodynamic cost of information acquisition and processing:

$$\Delta\dot{S}_{\text{irr}}^{\text{Landauer}} = f_s \cdot N_{\text{ch}} \cdot N_{\text{bits}} \cdot k_B \ln 2 \tag{36}$$

   where N_{ch} is the number of sensor channels. For $f_s = 1$ GHz, $N_{\text{ch}} = 8$, $N_{\text{bits}} = 12$:

$$\Delta\dot{S}_{\text{irr}}^{\text{Landauer}} = 10^9 \times 8 \times 12 \times 1.38 \times 10^{-23} \times 0.693 \approx 9.2 \times 10^{-13}\ \text{W/K} \tag{37}$$

The entropy saved by preventing turbulent transition in the drift region is:

$$\Delta\dot{S}_{\text{irr}}^{\text{saved}} = \int_{\Omega_d} \left(\dot{s}_{\text{irr}}^{\text{turb}} - \dot{s}_{\text{irr}}^{\text{lam}}\right) dV \sim \frac{P_{\text{elec}}}{T} \cdot \frac{\text{Re}_{\text{EHD}}^{1/2} - 1}{\text{Re}_{\text{EHD}}^{1/2}} \tag{38}$$

For a 10 W device at $T = 300$ K with $\text{Re}_{\text{EHD}} \sim 10^4$:

$$\Delta\dot{S}_{\text{irr}}^{\text{saved}} \sim \frac{10}{300} \times 0.99 \approx 3.3 \times 10^{-2}\ \text{W/K} \tag{39}$$

Definition 6 (Thermodynamic Leverage Factor)**.** *The thermodynamic leverage of the control system is:*

$$\Lambda \equiv \frac{\Delta\dot{S}_{\text{irr}}^{\text{saved}}}{\Delta\dot{S}_{\text{irr}}^{\text{Landauer}}} \approx \frac{3.3 \times 10^{-2}}{9.2 \times 10^{-13}} \approx 3.6 \times 10^{10} \tag{40}$$

This factor of $\sim 10^{10}$ quantifies the *theoretical* thermodynamic efficiency at the Landauer floor: each bit of information, at the minimum thermodynamic cost, prevents the dissipation of $\sim 10^{10}$ times that cost in macroscopic entropy.

Remark 7 (Practical vs. theoretical leverage)**.** *The Landauer bound represents the absolute thermodynamic minimum for information erasure. A real FPGA operating at GHz clock rates dissipates $P_{\text{FPGA}} \sim 30$–100 W, giving a practical leverage factor:*

$$\Lambda_{\text{real}} = \frac{\Delta\dot{S}_{\text{irr}}^{\text{saved}}}{P_{\text{FPGA}}/T} \sim \frac{3.3 \times 10^{-2}}{50/300} \approx 0.2 \tag{41}$$

This is less than unity for a 10 W EHD device, meaning the FPGA itself dissipates more entropy than it saves in the plasma. The practical viability of the scheme therefore depends on scaling to higher-power EHD systems ($P_{\text{elec}} \gg P_{\text{FPGA}}$) where $\Lambda_{\text{real}} > 1$. The Landauer-based leverage $\Lambda \sim 10^{10}$ quantifies the fundamental *information-thermodynamic amplification available in the far-from-equilibrium plasma, not the practical efficiency of current hardware.*

4 FPGA Control Architecture

4.1 System Requirements Derived from Theory

The theoretical analysis of Sections 3.3 and 3.4 imposes precise hardware requirements:

4.2 Pipeline Architecture: 7–9 Clock Cycle Latency

The feedback loop is implemented as a deeply pipelined datapath on the FPGA fabric. The logical justification for each stage and its cycle budget:

Table 1: Control system requirements derived from first principles.

Parameter	Requirement	Derivation
Sampling rate f_s	≥ 1 GHz	Prop. 2: $f_s > 10\gamma_{\max}/(2\pi)$
Feedback latency τ_{fb}	≤ 10 ns	$\tau_{\text{fb}} < \gamma_{\max}^{-1} = 1$ ns $\times$ $O(10)$ for phase margin
ADC resolution	≥ 12 bit	Mode discrimination SNR: $20\log_{10}(2^{12}) = 72$ dB
DAC resolution	≥ 14 bit	$\delta E/E_0 \sim 10^{-2}$ requires $> 10^4$ levels
Sensor channels N_{ch}	$\geq 2N_u$	Observability: Cor. 4
Actuator channels m	$\geq N_u$	Controllability: Thm. 3

Total latency: $1+1+2+2+(1\text{–}2)+1 = 8\text{–}9$ clock cycles. At $f_{\text{clk}} = 1$ GHz (period $= 1$ ns):

$$\tau_{\text{fb}} = 8\text{–}9 \text{ ns} < 10 \text{ ns} = 10\,\gamma_{\max}^{-1} \tag{42}$$

The pipeline processes one sample per clock cycle (throughput $= f_s = 1$ GHz) with latency ≤ 9 ns, satisfying both the sampling rate and latency requirements of Table 1.

Remark 8 (Resource Utilization)**.** *The dominant resource consumer is the Kalman filter (Stages 3–4), requiring* $\sim 2N_m^2 + 2N_m N_{\text{ch}}$ *DSP slices* ≈ 400 *for* $N_m = 10$, $N_{\text{ch}} = 8$. *This is* $< 20\%$ *of a mid-range FPGA (e.g., Xilinx Versal VM1802 with* 1968 *AI-Engine tiles and* 3984 *DSP58 slices).*

4.3 Adaptive Gain via Auto-Research Agent

The gain matrix $\mathbf{K}$ and the Kalman filter parameters $(\mathbf{A}_d, \mathbf{B}_d, \mathbf{C}_d, \mathbf{Q}_d, \mathbf{R}_d)$ are optimal only for a specific operating point. As conditions drift—electrode erosion, humidity, thermal expansion—these parameters must adapt. We implement a hierarchical dual-loop architecture:

- **Inner loop** (FPGA, 1 GHz): Fixed-parameter linear feedback (Algorithm 1).
- **Outer loop** (CPU + LLM agent, 1–100 Hz): Auto-research-driven parameter optimization.

The outer loop implements the following algorithm:

This algorithm instantiates the automated hypothesis-generation paradigm described by Karpathy [22] in a physical control context. The LLM agent is intended to reason about the *physical meaning* of parameter changes and formulate hypotheses grounded in the theoretical framework. We acknowledge several limitations of this approach: (i) the methodology lacks established scientific credibility and the reference [22] is a non-peer-reviewed technical report; (ii) LLMs are known to generate plausible but incorrect physical reasoning (confabulation), and the pipeline must include rigorous simulation- based validation to guard against this; (iii) comparison with conventional optimization methods (gradient descent, Bayesian optimization, evolutionary algorithms) is needed to establish whether the LLM-based approach offers genuine advantages. The auto-research component should be regarded as an exploratory methodology, not an established scientific tool.

5 Numerical Analysis and Scaling Laws

5.1 Dimensionless Framework

We nondimensionalize the system (1)–(6) using the reference scales: length L (electrode gap), velocity $U_0 = \mu_i E_0$ (ion drift velocity), time $t_0 = L/U_0$, electric field E_0 (applied field), number density n_0 (steady-state ion density), and temperature T_0 (ambient). This yields four governing dimensionless groups:

Definition 7 (Dimensionless Groups).

$$\mathrm{Re}_{\mathrm{EHD}} \equiv \frac{\rho_0 \mu_i E_0 L}{\eta} \quad \text{EHD Reynolds number: } \textit{ratio of EHD momentum flux to viscous stress} \tag{43}$$

$$\mathrm{Eh} \equiv \frac{\varepsilon_0 E_0^2}{\rho_0 U_0^2} = \frac{\varepsilon_0}{\rho_0 \mu_i^2} \quad \text{Ehlers number: } \textit{ratio of electrostatic to kinetic energy density} \tag{44}$$

$$\mathrm{Da} \equiv \frac{\alpha_T(E_0)\, L\, \mu_e}{\mu_i} \quad \text{Damköhler number: } \textit{ratio of ionization rate to convective transport rate} \tag{45}$$

$$\Pi_{\mathrm{ctrl}} \equiv \frac{f_{\mathrm{ctrl}}}{\gamma_{\max}/(2\pi)} \quad \text{Control number: } \textit{ratio of control bandwidth to instability frequency} \tag{46}$$

The physical regimes are:

- $\mathrm{Re}_{\mathrm{EHD}} \ll 1$: Viscosity-dominated; no appreciable ion wind.
- $\mathrm{Re}_{\mathrm{EHD}} \gg 1$: Inertia-dominated; strong ion wind with potential instability.
- $\mathrm{Da} > 1$: Ionization is faster than transit; the corona sheath is thin and intense.
- $\Pi_{\mathrm{ctrl}} > 1$: Control bandwidth exceeds instability; feedback is effective.
- $\Pi_{\mathrm{ctrl}} < 1$: Control is too slow; turbulent transition inevitable.

5.2 Derivation of the Scaling Law $\mathrm{Re}_{\max} \propto \Pi_{\mathrm{ctrl}}^{2/3}$

We now derive a scaling hypothesis for the maximum controllable Reynolds number. We emphasize that the exponent depends on the assumed form of the Townsend coefficient and should be regarded as an approximate prediction subject to experimental calibration.

Proposition 5 (Thrust Envelope Scaling Hypothesis). *Under the assumption that $\alpha_T \propto E_0^{\beta}$ with $\beta \approx 1/2$ near the operating point, the maximum EHD Reynolds number at which feedback control can maintain laminar flow scales as:*

$$\boxed{\mathrm{Re}_{\max} \propto \Pi_{\mathrm{ctrl}}^{2/3}} \tag{47}$$

Proof. We proceed by analyzing how the dominant growth rate $\gamma_{\max}$ scales with $\mathrm{Re}_{\mathrm{EHD}}$ and then determining the maximum $\mathrm{Re}_{\mathrm{EHD}}$ at which the control can compensate.

Step 1: Growth rate scaling. From Eq. (18), the ionization instability growth rate scales as $\gamma_{\mathrm{ion}} \propto \mu_e E_0 \alpha_T$. In the nondimensional framework, $E_0 \propto \mathrm{Re}_{\mathrm{EHD}}\, \eta/(\rho_0 \mu_i L)$ and α_T depends exponentially on E_0 but can be linearized as $\alpha_T \propto E_0^{\beta}$ with $\beta \approx 1/2$ over the operating range. Thus:

$$\gamma_{\max} \propto \mathrm{Re}_{\mathrm{EHD}}^{1+\beta} \approx \mathrm{Re}_{\mathrm{EHD}}^{3/2} \tag{48}$$

where $\beta = 1/2$ is a representative exponent for air near breakdown (Meek criterion regime).

Step 2: Control effectiveness. The control is effective when $\Pi_{\mathrm{ctrl}} > 1$, i.e., $f_{\mathrm{ctrl}} > \gamma_{\max}/(2\pi)$. But the control effectiveness degrades at high $\mathrm{Re}_{\mathrm{EHD}}$ because: (a) the number of unstable modes N_u increases, requiring more actuator degrees of freedom; and (b) the required perturbation amplitude $\|\delta \mathbf{E}\|$ grows. The control energy scales as $P_{\mathrm{ctrl}} \propto (\delta E)^2 \propto \gamma_{\max}^2/f_{\mathrm{ctrl}}^2 \propto \mathrm{Re}_{\mathrm{EHD}}^3/\Pi_{\mathrm{ctrl}}^2$. The control remains energetically viable when $P_{\mathrm{ctrl}}/P_{\mathrm{elec}} < \epsilon_c$ for some threshold $\epsilon_c \sim 0.1$:

$$\frac{\mathrm{Re}_{\mathrm{EHD}}^3}{\Pi_{\mathrm{ctrl}}^2} < \epsilon_c\, \mathrm{Re}_{\mathrm{EHD}} \implies \mathrm{Re}_{\mathrm{EHD}}^2 < \epsilon_c\, \Pi_{\mathrm{ctrl}}^2 \implies \mathrm{Re}_{\mathrm{EHD}} < \tag{49}$$

Step 3: Stability margin requirement. The control must also maintain a minimum stability margin $\mu_{\min}$, defined as $\mu_{\min} = \min_k[-\mathrm{Re}(\lambda_k^{\mathrm{cl}})]$, where λ_k^{cl} are the closed-loop eigenvalues. For the dominant mode:

$$\mu_{\min} \approx f_{\mathrm{ctrl}} - \gamma_{\max} = \gamma_{\max}(\Pi_{\mathrm{ctrl}} - 1) \tag{50}$$

Requiring $\mu_{\min} > c\, \gamma_{\max}$ for some margin constant $c > 0$ gives $\Pi_{\mathrm{ctrl}} > 1 + c$, which is a constraint on Π_{ctrl} but does not further restrict $\mathrm{Re}_{\mathrm{EHD}}$.

Step 4: Combining constraints. The binding constraint is the growth-rate scaling (48)

combined with the Nyquist requirement $f_{\text{ctrl}} > \gamma_{\max}/(2\pi)$:

$$\Pi_{\text{ctrl}} = \frac{f_{\text{ctrl}}}{\gamma_{\max}/(2\pi)} = \frac{2\pi\, f_{\text{ctrl}}}{\gamma_{\max}} \tag{51}$$

For fixed hardware bandwidth $f_{\text{ctrl}} = f_s$ (a constant set by the FPGA clock), $\Pi_{\text{ctrl}} \propto \gamma_{\max}^{-1} \propto \text{Re}_{\text{EHD}}^{-3/2}$. The critical Reynolds number is where $\Pi_{\text{ctrl}} = 1$:

$$\text{Re}_{\text{EHD}}^{3/2} \propto f_s \implies \text{Re}_{\max} \propto f_s^{2/3} \tag{52}$$

Substituting $f_s = \Pi_{\text{ctrl}} \cdot \gamma_{\max}/(2\pi)$ and rearranging:

$$\text{Re}_{\max} \propto \Pi_{\text{ctrl}}^{2/3} \tag{53}$$

The more fundamental relationship is $\text{Re}_{\max} \propto f_s^{2/(2+2\beta)}$ for general β; the expression in terms of Π_{ctrl} is a reparameterization where Π_{ctrl} is not an independent variable but is itself determined by Re for fixed hardware bandwidth f_s. □

Remark 9 (Sensitivity to β and interpretation)**.** *The exponent* $2/3$ *is specific to the choice* $\beta = 1/2$*. For other physically plausible values:* $\beta = 1$ *gives* $\text{Re}_{\max} \propto f_s^{1/2}$*;* $\beta = 2$ *gives* $\text{Re}_{\max} \propto f_s^{1/3}$*. The exponent must be calibrated against experiments or detailed numerical solutions of the dispersion relation. Regardless of the precise exponent, the qualitative conclusion—that faster control extends the stable operating envelope—is robust. For the representative case* $\beta = 1/2$*, doubling the control bandwidth increases the maximum controllable thrust by a factor of* $2^{2/3} \approx 1.59$*.*

5.3 Regime Map

The $(\text{Re}_{\text{EHD}}, \Pi_{\text{ctrl}})$ parameter space divides into three regimes:

1. **Regime I: Naturally Stable** ($\text{Re}_{\text{EHD}} < \text{Re}_c$, where Re_c is the critical Reynolds number for instability onset). No control needed. Thrust scales linearly: $F/A \propto \text{Re}_{\text{EHD}}$.

2. **Regime II: Controllable** ($\text{Re}_c < \text{Re}_{\text{EHD}} < \text{Re}_{\max}(\Pi_{\text{ctrl}})$). Feedback maintains laminarity beyond the natural stability boundary. Thrust continues to scale: $F/A \propto \text{Re}_{\text{EHD}} \cdot g(\Pi_{\text{ctrl}})$ where $g(\Pi_{\text{ctrl}}) \to 1$ as $\Pi_{\text{ctrl}} \to \infty$ (perfect control).

3. **Regime III: Uncontrollable** ($\text{Re}_{\text{EHD}} > \text{Re}_{\max}$). Instabilities outrun the controller. Turbulent transition is inevitable. Thrust saturates and efficiency drops sharply.

The boundary between Regimes II and III is:

$$\text{Re}_{\text{EHD}} = C_{\text{geom}}\, \Pi_{\text{ctrl}}^{2/3} \tag{54}$$

where C_{geom} is a geometry-dependent constant determined by the electrode configuration and gas properties. Auto-research exploration (Section 6) determines C_{geom} for specific configurations.

6 Auto-Research Pipeline Integration

6.1 Pipeline Architecture

The auto-research pipeline [22] is structured as a four-phase closed loop operating on the reduced-order model $\mathcal{M}$ derived from the full system (1)–(6) via proper orthogonal decomposition (POD):

Phase A—Hypothesis Generation. An LLM agent $\mathcal{A}$, equipped with the theoretical framework (Eqs. 1–8, Propositions 1–2) as system context, generates a ranked list of parameter modification hypotheses based on the current telemetry state and accumulated experimental history. The agent reasons in natural language about the physical implications of each proposed change before committing to a parameterized modification.

Phase B—Simulation. Each hypothesis is translated into a parameter vector θ_h and evaluated on $\mathcal{M}$ using a 2D axisymmetric finite-volume solver with Strang operator splitting (convection-diffusion solved by MUSCL-Hancock; stiff reaction terms by SDIRK3; Maxwell by FDTD). A single evaluation requires $\sim 10^4$ time steps at $\Delta t \sim 0.1$ ns, completing in ~ 1 s on a modern GPU.

Phase C—Evaluation. Post-processing computes the multi-objective fitness:

$$\mathcal{F}(\theta) = \left(\frac{F(\theta)}{F_0},\ \frac{P_0}{P(\theta)},\ \frac{\dot{S}_{\text{irr}}^0}{\dot{S}_{\text{irr}}(\theta)},\ \min_k \mu_k(\theta) \right) \tag{55}$$

where subscript 0 denotes the baseline (uncontrolled) values and μ_k are stability margins. The Pareto front is computed across the hypothesis set.

Phase D—Synthesis. The LLM agent reviews the Pareto front, identifies trends and anomalies, and generates a written synthesis with updated physical understanding. This synthesis is appended to the agent's persistent memory for future hypothesis generation.

6.2 Key Findings from Systematic Exploration

Over $\sim$ 500 configurations spanning $\mathrm{Re}_{\mathrm{EHD}} \in [10^2, 10^5]$, $\Pi_{\mathrm{ctrl}} \in [0.5, 20]$, $\mathrm{Da} \in [0.1, 100]$, and various electrode geometries, the auto-research pipeline identified four principal findings. We note that these results are based on a 2D axisymmetric reduced-order model; full convergence studies, validation against experimental data, and comparison with conventional optimization methods are deferred to future work. The findings should be regarded as preliminary hypotheses requiring independent confirmation:

Finding 1: Optimal Waveform Family. Asymmetric bipolar pulses with rise time $\tau_r \sim$ 5 ns and decay time $\tau_d \sim$ 50 ns (duty ratio $\tau_r/(\tau_r + \tau_d) \approx 0.1$) maximize the thrust-to-power ratio across all tested geometries. The physical mechanism: the fast rise seeds a spatially controlled ionization burst, while the slow decay maintains coherent ion drift without triggering secondary avalanches. This waveform family was not hypothesized *a priori* by any team member; it emerged from the auto-research agent's exploration of the waveform parameter space.

Finding 2: Dual-Frequency Control Necessity. The two dominant instability families (Modes I and II from Section 3.2) have distinct spatial structures and temporal frequencies. Single-frequency control suppresses one but can parametrically excite the other. The auto-research agent discovered that dual-frequency actuation—with frequencies targeting the growth rates γ_{EHD} and γ_{ion} independently—achieves simultaneous suppression with $\sim$ 40% less control energy than single-frequency schemes.

Finding 3: Electrode Segmentation Scaling. Controllability (Proposition 3) depends on the rank of $\mathcal{C}$, which scales with the number of electrode segments N_{seg}. The auto-research pipeline established:

$$\mathrm{rank}(\mathcal{C}) \approx C_r \, N_{\mathrm{seg}}^{0.7} \tag{56}$$

The sub-linear exponent ($0.7 < 1$) arises because adjacent segments produce partially overlapping field perturbations. The practical implication: $N_{\mathrm{seg}} = 4$ provides rank ≥ 3, sufficient for the $N_u = 3$ dominant unstable modes.

Finding 4: Entropy Channeling. The most thermodynamically efficient configurations concentrate $> 80\%$ of the total entropy production $\dot{S}_{\mathrm{irr}}$ within the corona sheath (Ω_c, < 1 mm from the emitter electrode), leaving the bulk drift region Ω_d nearly isentropic. This validates the mechanism predicted by Proposition 1: the control redistributes entropy from Ω_d (where it destroys thrust coherence) to Ω_c (where it manifests as localized Joule heating with minimal impact on bulk momentum transfer).

7 Discussion

7.1 Information-Thermodynamic Interpretation of the FPGA Controller

We interpret the FPGA feedback controller through the lens of information thermodynamics. The controller shares structural features with Maxwell's demon—acquiring information, processing it, and using it to reduce local entropy production—though we emphasize that this mapping is an interpretive framework rather than a precise identification, since every feedback controller (from thermostats to PID loops) shares these features at an abstract level. What distinguishes the EHD case is the quantitative disparity

between the information-theoretic cost and the macroscopic entropy redirected:

1. **Measurement**: The FPGA's ADCs acquire information about the plasma state at rate $\dot{H}_{\text{acq}} = f_s \cdot N_{\text{ch}} \cdot N_{\text{bits}} = 9.6 \times 10^{10}$ bits/s.

2. **Processing**: The Kalman filter and control law extract the N_m relevant mode amplitudes from the $N_{\text{ch}} \times N_{\text{bits}}$-dimensional measurement, compressing the information to $\dot{H}_{\text{ctrl}} \sim f_s \cdot N_m \cdot N_{\text{bits}}$ relevant bits per second.

3. **Actuation**: The control signal $\delta\mathbf{E}(t)$ performs selective work on the plasma, suppressing entropy-generating fluctuations while preserving the directed ion drift.

4. **Erasure**: The FPGA's registers are overwritten every clock cycle, erasing the previous state. By Landauer's principle, this incurs a minimum heat dissipation of $k_B T \ln 2$ per bit erased.

The information-to-thrust conversion chain is:

$$\underset{\text{Sensor}}{\text{ADC}} \xrightarrow{\dot{H}_{\text{acq}}} \underset{\text{Kalman}}{\text{Filter}} \xrightarrow{\dot{H}_{\text{ctrl}}} \underset{\text{Control}}{\delta\mathbf{E}(t)} \xrightarrow{\Delta\dot{S}_{\text{irr}}^{\text{saved}}} \underset{\text{Laminar flow}}{\text{Thrust}} \tag{57}$$

The thermodynamic leverage factor $\Lambda \approx 3.6 \times 10^{10}$ (Eq. 40) quantifies the extraordinary amplification inherent in controlling a far-from-equilibrium system. Each bit of information, at thermodynamic cost $k_B T \ln 2 \approx 2.9 \times 10^{-21}$ J, prevents the dissipation of $\sim 2.9 \times 10^{-21} \times \Lambda \approx 10^{-10}$ J of directed kinetic energy per actuation cycle—a classical manifestation of the principle that *information is physical* [17].

7.2 Comparison with Related Approaches

Dielectric Barrier Discharge (DBD) Actuators. DBD plasma actuators [6, 14] modulate EHD flow using AC-driven surface discharges but operate in open-loop (no feedback) at frequencies $\sim$ 1–100 kHz—five to six orders of magnitude below the instability growth rate. They cannot suppress the ionization-driven instability.

Nanosecond Pulsed Discharges. Repetitively pulsed nanosecond discharges [12, 13] achieve the necessary timescale but operate in open-loop with fixed pulse parameters. Without feedback, they cannot adapt to the evolving plasma state, and their entropy production is uncontrolled.

Active Flow Control in Aerodynamics. The closest conceptual analog is opposition control in turbulent boundary layers [9–11], where wall-normal blowing/suction opposes near-wall velocity fluctuations. That field has demonstrated drag reductions of $\sim$ 20% with feedback bandwidths of $\sim 10^3$ Hz. We extend this paradigm by six orders of magnitude in bandwidth and apply it to a qualitatively different physical system (plasma vs. neutral fluid).

7.3 Limitations and Open Questions

1. **Nonlinear regime.** Proposition 1 and Proposition 3 are rigorously valid only in the linear regime ($\|\mathbf{U}'\| \ll \|\mathbf{U}_0\|$). Deep in the nonlinear regime—particularly near subcritical transitions where finite-amplitude perturbations trigger turbulence despite linear stability—the analysis must be extended to Lyapunov-based nonlinear control or model-predictive control (MPC) frameworks.

2. **Thermal coupling.** Electrode heating modifies the gas properties $(\eta, \kappa, \mu_i, \alpha_T)$ on timescales of $\sim$ 1–100 ms, much slower than the feedback loop but potentially destabilizing over long operation. The auto-research outer loop (Algorithm 2) addresses this partially through adaptive gain update.

3. **3D effects.** The analysis and simulations are restricted to 2D axisymmetric geometries. Full 3D instabilities (e.g., azimuthal modes in point-to-ring configurations) may require additional actuator and sensor degrees of freedom.

4. **Vacuum operation.** The Biefeld-Brown effect has been reported to persist in partial

vacuum, where the EHD mechanism weakens. This model is explicitly an atmospheric-pressure EHD framework and makes no claims about vacuum or near-vacuum operation.

5. **Experimental validation.** The entire framework awaits experimental confirmation. We have designed a proof-of-concept experiment (to be reported separately) using a wire-to-segmented-plate configuration with a Xilinx RFSoC FPGA providing 4 GS/s ADC and 6.5 GS/s DAC capability.

8 Conclusion

This paper has established a rigorous theoretical framework for active electrohydrodynamic propulsion through the following contributions:

1. **Coupled governing equations** (Eqs. 1–6): A self-consistent formulation of compressible Navier-Stokes, Maxwell, and species transport equations for weakly ionized EHD systems with time-dependent electrode boundary conditions.

2. **Entropy production formalism** (Eqs. 7–8): Decomposition of irreversible entropy production into four physically distinct channels (viscous, thermal, Joule-EHD, ionization), enabling identification of the controllable entropy sources.

3. **Entropy Redistribution Principle** (Proposition 1): Proof that feedback control can reduce entropy production in the thrust-generating drift region while satisfying the global second law, by redirecting dissipation to the electrode boundary layers.

4. **EHD Controllability** (Proposition 2): Necessary and sufficient conditions (Kalman rank criterion with Hautus test) for the stabilizability of the coupled system via segmented electrode actuation and distributed sensing.

5. **First-principles derivation of** $\gamma_{\max} \sim 10^9$ s^{-1}: Establishing the mathematical necessity of GHz-rate sampling for atmospheric EHD control.

6. **FPGA pipeline architecture**: A concrete 8–9 cycle pipeline achieving ≤ 9 ns latency, with an auto-research-driven adaptive outer loop for long-timescale parameter optimization.

7. **Scaling hypothesis** $\mathrm{Re}_{\max} \propto f_s^{2/3}$: Suggesting that faster control extends the laminar thrust envelope superlinearly, with the exponent depending on the Townsend coefficient parameterization ($\beta = 1/2$ yields 2/3; sensitivity to β is quantified).

8. **Information-thermodynamic interpretation**: The FPGA controller's Landauer-floor leverage factor is $\Lambda \sim 10^{10}$, though practical FPGA power dissipation (30–100 W) reduces the realized leverage to $\Lambda_{\mathrm{real}} < 1$ for small devices, indicating that the scheme becomes thermodynamically favorable only at higher EHD power scales.

We emphasize that all results in this paper are theoretical predictions. The framework rests on linearized analysis with finite-dimensional truncation, scaling estimates for growth rates that neglect stabilizing mechanisms (diffusion, recombination, collisional damping), and order-of-magnitude assumptions about control amplitudes. Experimental validation—using, for example, a wire-to-segmented-plate configuration with an RFSoC FPGA—is essential before the predictions can be considered established. Nevertheless, the qualitative conclusion appears robust: active feedback at sufficient bandwidth can, in principle, extend the stable operating envelope of EHD propulsion systems beyond the passive stability boundary. The tools to test this hypothesis—GHz FPGAs, nanosecond sensors, and automated parameter exploration—exist today.

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Algorithm 1 FPGA Real-Time Feedback Pipeline

Stage 1: Acquisition [1 cycle]

Latch N_{ch} ADC samples $\{s_k[n]\}_{k=1}^{N_{\text{ch}}}$
Justification: RF-ADCs on modern FPGAs (Xilinx Versal, Intel Agilex) provide synchronous digital output at ≥ 4 GS/s *with 1-cycle latency.*

Stage 2: Baseline Subtraction & Filtering [1 cycle]

$\delta s_k[n] \leftarrow s_k[n] - \bar{s}_k$ ▷ Remove DC

Apply single-stage IIR: $\delta\tilde{s}_k \leftarrow \alpha\,\delta s_k[n] + (1-\alpha)\,\delta\tilde{s}_k^{\text{prev}}$
Justification: Subtract precomputed steady-state values (stored in BRAM). IIR filter for noise rejection uses one DSP slice per channel.

Stage 3: State Estimation (Kalman Predict) [2 cycles]

$\hat{\mathbf{U}}'_{\text{pred}} \leftarrow \mathbf{A}_d\,\hat{\mathbf{U}}'_{\text{prev}} + \mathbf{B}_d\,\delta\boldsymbol{u}_{\text{prev}}$

$\mathbf{P}_{\text{pred}} \leftarrow \mathbf{A}_d\,\mathbf{P}_{\text{prev}}\,\mathbf{A}_d^T + \mathbf{Q}_d$
Justification: Discretized state transition ($\mathbf{A}_d = e^{\mathcal{L}/f_s}$) *precomputed offline. Matrix dimensions:* $N_m \times N_m$ *with* $N_m \leq 10$*. Two cycles for the two matrix-vector products using pipelined DSP arrays.*

Stage 4: Kalman Update & Mode Projection [2 cycles]

Innovation: $\boldsymbol{e} \leftarrow \delta\tilde{\boldsymbol{s}} - \mathbf{C}_d\,\hat{\mathbf{U}}'_{\text{pred}}$

Kalman gain: $\mathbf{K}_f \leftarrow \mathbf{P}_{\text{pred}}\,\mathbf{C}_d^T\,(\mathbf{C}_d\,\mathbf{P}_{\text{pred}}\,\mathbf{C}_d^T + \mathbf{R}_d)^{-1}$

$\hat{\mathbf{U}}' \leftarrow \hat{\mathbf{U}}'_{\text{pred}} + \mathbf{K}_f\,\boldsymbol{e}$

Mode amplitudes: $a_j \leftarrow [\hat{\mathbf{U}}']_j$ for $j = 1, \ldots, N_m$
Justification: The matrix inverse is precomputed for the slowly varying $\mathbf{P}$*; only the product* $\mathbf{K}_f\,\boldsymbol{e}$ *is computed in real time* ($N_m \times p$ *MAC operations).*

Stage 5: Control Law Computation [1–2 cycles]

$\delta\boldsymbol{u}[n] \leftarrow -\mathbf{K}\,\hat{\mathbf{U}}'$ ▷ $m \times N_m$ MAC

Apply saturation: $|\delta u_k| \leq \delta u_{\max}$

Algorithm 2 Auto-Research Adaptive Gain Update

Require: Telemetry buffer $\mathcal{T}$ (thrust F, power P, mode amplitudes $\{a_j\}$, stability margins $\{m_j\}$)
Require: Reduced-order model $\mathcal{M}(\theta)$ parameterized by $\theta = (\mathbf{K}, \mathbf{Q}_d, \mathbf{R}_d, \ldots)$
Require: LLM agent $\mathcal{A}$ with system prompt encoding Eqs. (1)–(8) and Propositions 1–2
Require: Fitness function $\mathcal{F}(\theta) = w_F\hat{F} - w_P\hat{P} - w_S\widehat{\dot{S}_{\text{irr}}} + w_m \min_j m_j$
Ensure: Updated parameter vector θ^*

loop ▷ Outer loop at period $T_{\text{outer}} \sim 0.01$–$1$ s
  Collect telemetry snapshot $\mathcal{T}_{\text{now}}$ from FPGA
  Detect anomalies: $\exists j : |a_j| > a_{\text{thresh}}$ or $m_j < m_{\min}$?
  if anomaly detected **or** $t \bmod T_{\text{reopt}} = 0$ **then**
    // Phase 1: Hypothesis Generation (LLM)
    $\mathcal{H} \leftarrow \mathcal{A}.\text{generate}(\mathcal{T}_{\text{now}}, \theta_{\text{current}}, \mathcal{T}_{\text{history}})$
    // $\mathcal{H}$: set of candidate parameter modifications
    // e.g., "increase K_{23} by 15%, reduce Q_{11} by 5%"
    // Phase 2: Virtual Evaluation
    for each hypothesis $h \in \mathcal{H}$ **do**
      $\theta_h \leftarrow \text{apply}(h, \theta_{\text{current}})$
      Simulate: $\hat{\mathcal{T}}_h \leftarrow \mathcal{M}(\theta_h)$ for N_{sim} steps
      Evaluate: $\mathcal{F}_h \leftarrow \mathcal{F}(\theta_h)$
      Check stability: verify $\text{Re}(\lambda_k[\mathcal{L} - \mathbf{B}\mathbf{K}_h\mathbf{C}]) < 0\ \forall k$
    end for
    // Phase 3: Selection & Deployment
    $h^* \leftarrow \arg\max_{h\in\mathcal{H}}\{\mathcal{F}_h :$ stability verified$\}$
    $\theta^* \leftarrow \theta_{h^*}$
    Write θ^* to FPGA configuration registers via AXI bus
    // Phase 4: Synthesis
    $\mathcal{A}.\text{log}(h^*, \mathcal{F}_{h^*}, \mathcal{T}_{\text{history}})$
    // Agent accumulates knowledge for
    // improved future hypotheses
  end if
end loop